

Understanding Printer Command Codes (ESC POS)

The printer codes that we use in manager console, known as ESC POS codes, are decimally-represented ASCII characters that are combined to make Epson Print Commands. We use Epson commands by default because SNBC printers are Epson-emulating printers. These codes can be used for other Epson-emulating printers as well.

*These codes are the main reason we don't use the USB interface for printers - the software hasn't been designed to. These codes are communicated via serial and ethernet, but any USB printer would have to be implemented via a virtual com port.

The tools you will need to understand this KB are:

The Epson ESC POS code list:

https://download4.epson.biz/sec_pubs/pos/reference_en/escpos/commands.html

An ASCII conversion chart, like the one below

Decimal - Binary - Octal - Hex – ASCII Conversion Chart

Decimal	Binary	Octal	Hex	ASCII	Decimal	Binary	Octal	Hex	ASCII	Decimal	Binary	Octal	Hex	ASCII	Decimal	Binary	Octal	Hex	ASCII
0	00000000	000	00	NUL	32	00100000	040	20	SP	64	01000000	100	40	@	96	01100000	140	60	`
1	00000001	001	01	SOH	33	00100001	041	21	!	65	01000001	101	41	A	97	01100001	141	61	a
2	00000010	002	02	STX	34	00100010	042	22	"	66	01000010	102	42	B	98	01100010	142	62	b
3	00000011	003	03	ETX	35	00100011	043	23	#	67	01000011	103	43	C	99	01100011	143	63	c
4	00000100	004	04	EOT	36	00100100	044	24	\$	68	01000100	104	44	D	100	01100100	144	64	d
5	00000101	005	05	ENQ	37	00100101	045	25	%	69	01000101	105	45	E	101	01100101	145	65	e
6	00000110	006	06	ACK	38	00100110	046	26	&	70	01000110	106	46	F	102	01100110	146	66	f
7	00000111	007	07	BEL	39	00100111	047	27	'	71	01000111	107	47	G	103	01100111	147	67	g
8	00001000	010	08	BS	40	00101000	050	28	(	72	01001000	110	48	H	104	01101000	150	68	h
9	00001001	011	09	HT	41	00101001	051	29	)	73	01001001	111	49	I	105	01101001	151	69	i
10	00001010	012	0A	LF	42	00101010	052	2A	*	74	01001010	112	4A	J	106	01101010	152	6A	j
11	00001011	013	0B	VT	43	00101011	053	2B	+	75	01001011	113	4B	K	107	01101011	153	6B	k
12	00001100	014	0C	FF	44	00101100	054	2C	,	76	01001100	114	4C	L	108	01101100	154	6C	l
13	00001101	015	0D	CR	45	00101101	055	2D	-	77	01001101	115	4D	M	109	01101101	155	6D	m
14	00001110	016	0E	SO	46	00101110	056	2E	.	78	01001110	116	4E	N	110	01101110	156	6E	n
15	00001111	017	0F	SI	47	00101111	057	2F	/	79	01001111	117	4F	O	111	01101111	157	6F	o
16	00010000	020	10	DLE	48	00110000	060	30	0	80	01010000	120	50	P	112	01110000	160	70	p
17	00010001	021	11	DC1	49	00110001	061	31	1	81	01010001	121	51	Q	113	01110001	161	71	q
18	00010010	022	12	DC2	50	00110010	062	32	2	82	01010010	122	52	R	114	01110010	162	72	r
19	00010011	023	13	DC3	51	00110011	063	33	3	83	01010011	123	53	S	115	01110011	163	73	s
20	00010100	024	14	DC4	52	00110100	064	34	4	84	01010100	124	54	T	116	01110100	164	74	t
21	00010101	025	15	NAK	53	00110101	065	35	5	85	01010101	125	55	U	117	01110101	165	75	u
22	00010110	026	16	SYN	54	00110110	066	36	6	86	01010110	126	56	V	118	01110110	166	76	v
23	00010111	027	17	ETB	55	00110111	067	37	7	87	01010111	127	57	W	119	01110111	167	77	w
24	00011000	030	18	CAN	56	00111000	070	38	8	88	01011000	130	58	X	120	01111000	170	78	x
25	00011001	031	19	EM	57	00111001	071	39	9	89	01011001	131	59	Y	121	01111001	171	79	y
26	00011010	032	1A	SUB	58	00111010	072	3A	:	90	01011010	132	5A	Z	122	01111010	172	7A	z
27	00011011	033	1B	ESC	59	00111011	073	3B	;	91	01011011	133	5B	[	123	01111011	173	7B	{
28	00011100	034	1C	FS	60	00111100	074	3C	<	92	01011100	134	5C	\	124	01111100	174	7C	
29	00011101	035	1D	GS	61	00111101	075	3D	=	93	01011101	135	5D	]	125	01111101	175	7D	}
30	00011110	036	1E	RS	62	00111110	076	3E	>	94	01011110	136	5E	^	126	01111110	176	7E	~
31	00011111	037	1F	US	63	00111111	077	3F	?	95	01011111	137	5F	_	127	01111111	177	7F	DEL

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ASCII Conversion Chart.doc Copyright © 2008, 2012 Donald Weiman 22 March 2012

Finally a onePOS installation to test with. For this KB, we will be referencing a standard SNBC thermal printer hardware setup.

As you can see from the hardware settings, commands are strung together like sentences where the spaces between the numbers are just like spaces between words.

To begin, let's convert the characters (the "words" in our command sentence) that are put together to make the "Normal Command" from decimal back to ASCII. Using the ASCII conversion chart, we can see that the decimal numbers convert as follows:

27 - ESC

33 - !

2 - STX

Referencing these ASCII characters to the Epson code list, we can see that we are using the "ESC !" command (further explained here: https://download4.epson.biz/sec_pubs/pos/reference_en/escpos/esc_exclamation.html).

Setup | Edit Hardware Device | Printer - Serial...

ID: 1 0.02452

UID: 1

Name: TERM 01-L0

Sort: 10

Device Type: Printer - Serial

Terminal: Terminal #01

Port: 1

Params: 115200,n,8,1

Redirect To: TERM 02-L0

Redirect Timeout: 45 Seconds

Normal Command: 27 33 2

Wide Command: 27 33 32

Red Command:

Exit Command: 10 10 10 10 10 27 105

Logo File:

Drawer 1: 27 112 0 100 10

Drawer 2: 27 112 1 100 10

Open Detect 1:

Open Detect 2:

Tall/Wide: 27 33 49 10

Bar Code: 29 104 50 29 107 65 11

Upside Begin:

Upside End:

1/2 Space:

Bitmap Line Size: 0

☐ Cut Redirected Chit
☐ HH Printer Target
☐ In-Active

OK

Cancel

Previous

Next

Last Update: 01-24-25 16:32

The "ESC !" command tells the printer that we are selecting "the character font and styles (emphasized, double-height, double-width, and underline) together..." In this command format "**27 33 n**", "27 33" represent the command itself and "**n**" represents what our desired outcome should be; in this case "2" emphasizes bold text.

Looking at the "Wide Command," we can see that we are still utilizing the "ESC !" command, but instead of default, we are using "32" which is defined in the command details in the decimal column, to the right of the "Double-Width Mode: On" function.

Moving on to the "Exit Command", the words we use in this sentence are:

10 - LF
27 - ESC
105 - i

Whereas "10" is the "Line Feed" function and "27 105" is the "ESC i" or "partial cut" function. The result of this command sentence is us telling the printer "after you print everything as defined above, feed the paper 5 times and cut the paper down to a point." That's how we get all of our chits to string together.

As you can see, sentences are made up entirely of commands or commands and their definable variables. The Drawer command is a pulse command (ESC p) with a formula that controls the specific drawer. The Epson page for the ESC P command (and other commands) lays out the formula in ASCII and decimal. The formula for our drawer in decimal is "27 112 **m t1 t2**" where "m" is the pin being pulsed to kick out the drawer and the pulse ON time is (t1 × 2 msec) and OFF time is (t2 × 2 msec).

For another example, some of our customers ask for the text to be bigger. Using the "ESC !" function, we could simply change the normal command to "27 33 24." This would make all of their text print bigger.

Following is a more advanced description of how you get the decimal code for multi-bit functions (where the 24 came from):

Bit	Decimal Value	Description
0	1	Select Font (0 = Font A, 1 = Font B)
1	2	Emphasized (bold)
2	4	Double-strike

Bit Decimal Value		Description
3	8	Double-height
4	16	Double-width
5	32	Reserved on official Epson docs (sometimes repurposed)
6	64	Reserved
7	128	Reserved

1. How Bits Add Up to Decimal

To find the decimal value of **n**, you **add** the decimal values of the individual bits you want to turn on:

- If you want to turn on bit 3 (Double-height), you add 8.
- If you also want to turn on bit 4 (Double-width), add 16.
- Together, bits 3 + 4 = 8 + 16 = 24.

Hence, **n = 24** (binary 0001 1000) means **double-height + double-width**.

Similarly:

- **n = 16** alone (binary 0001 0000) sets only bit 4 (Double-width).
- **n = 8** alone (binary 0000 1000) sets only bit 3 (Double-height).
- **n = 32** alone (binary 0010 0000) sets bit 5, which is *reserved* in the official spec but on many clone printers can mimic double-height or do something else.